KEYFACTOR

Post-Quantum Cryptography (PQC)

Preparing for the Next Cryptographic Transition

Mike Bodette
Senior Channel Solutions Engineer

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What is Post-Quantum Cryptography?

THE PROBLEM

Today's public-key cryptography depends on math that is hard for classical computers:

RSA depends on factoring large integers and ECC/ECDSA/ECDH depend on elliptic curve discrete logarithms

WHY QUANTUM BREAKS IT

Why quantum changes everything:

A sufficiently capable quantum computer could use Shor's algorithm to solve those problems efficiently, which would break the security foundation of RSA and ECC.

THE REPLACEMENT

Post-quantum cryptography uses new algorithms designed to resist attacks from both:
Classical computers as well as Quantum computers

ML-KEM
ML-DSA SLH-DSA

Why this matters to federal teams:

These are not research concepts anymore. They are now Federal Information Processing Standards, and NSA's CNSA 2.0 direction also requires quantum-resistant algorithms for National Security Systems.

ML-KEM (FIPS 203) for key establishment / key exchange

ML-DSA (FIPS 204) for digital signatures

SLH-DSA (FIPS 205) as an additional signature option

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Classical vs Quantum-Safe Cryptography

Today's Algorithms (Classical Cryptography)

- RSA → Based on integer factorization
- ECC → Based on elliptic curve discrete logarithms
- Security assumption: These problems are computationally infeasible

Quantum-Safe Algorithms (NIST PQC Standards)

- ML-KEM (FIPS 203) → Key encapsulation (replaces RSA/ECDH)
- ML-DSA (FIPS 204) → Digital signatures (replaces RSA/ECDSA)
- SLH-DSA (FIPS 205) → Hash-based signatures

Quantum Threat (Shor's Algorithm)

- Efficiently breaks RSA and ECC
- Turns "hard" math problems into solvable ones
- Impacts key exchange and digital signatures

Key Differences

- Much larger keys and signatures
- Different mathematical foundations (lattices, hash-based)
- Performance and compatibility tradeoffs

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Why this Transition is Different

Replaces the mathematical foundation of modern cryptography

Requires system-wide change, not incremental updates

Federally mandated timelines exist

Impacts performance, compatibility, and operations

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Why PQC Matters Today

Not Just in the Future

HARVEST NOW → ***DECRYPT LATER***

Quantum readiness assessments and migration planning are top of mind.

- **Long-lived data such as PII, IP, & regulated data is at risk**
- **Regulatory standards are being finalized and adopted**
- **Early experimentation by browsers, cloud providers, and CAs**
- **Applies to classified and long-lived federal data today**

The migration timeline is driven by data longevity, not quantum availability.

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Why PQC Matters Today

This is a coordinated federal push, not isolated guidance

NSM-10 is a U.S. federal directive that requires agencies to inventory cryptographic systems, assess the risks quantum computing poses to current encryption, and begin transitioning to quantum-resistant algorithms.

NIST has published PQC standards (FIPS 203, 204, 205) and recommends organizations begin migration planning now.

OMB M-23-02 mandates that federal agencies inventory cryptographic systems, identify quantum-vulnerable algorithms, and report annually on migration readiness.

CISA, NSA, and NIST jointly recommend organizations inventory cryptographic assets and develop PQC migration roadmaps.

NSA CNSA 2.0 mandates transition to quantum-resistant algorithms for national security systems, with defined timelines through 2035.

NIST SP 800-52 defines the federal TLS baseline today, while CAB Forum policy and PQC initiatives are driving the next evolution of digital trust.

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What Changes for Certificates and TLS in a Post-Quantum World

Today's TLS (Classical Model)

- Key Exchange: ECDHE or RSA
- Signatures: RSA or ECDSA
- Certificate Size: ~1-3 KB typical
- Handshake: Optimized, low latency

PQC TLS (What Actually Changes)

- **Key Exchange → KEM-Based**
  - ECDHE replaced or augmented by ML-KEM (Kyber)
  - Introduces encapsulation / decapsulation model
- **Signatures → PQC Algorithms**
  - RSA/ECDSA replaced or augmented by:
    - ML-DSA (Dilithium)
    - SLH-DSA (SPHINCS+)
- **Certificates**
  - Hybrid certificates (classical + PQC)
  - Larger public keys and signatures
  - Increased certificate chain size

Based on industry testing from Cloudflare, Google, and NIST PQC evaluations

Performance and Size Impact

- Key sizes: 2–10x larger than ECC
- Signatures: significantly larger (often KBs vs bytes)
- TLS handshake size: increases substantially
- Overhead is driven primarily by larger keys and handshake sizes, not just computation speed
- Hybrid mode introduces both classical and PQC operations in the same handshake

Observed Impact (Industry Benchmarks)

- Measured increases in CPU usage and handshake latency in hybrid PQC TLS deployments
- Higher memory and bandwidth utilization

Why this Matters Operationally

- More compute per connection
- More data transmitted per handshake
- Greater impact on:
  - APIs and microservices
  - High-throughput systems
  - Constrained environments (devices, edge)

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PQC Certificates in Practice

2026 Reality Check

WHAT'S POSSIBLE TODAY

EJBCA **CAN ISSUE…**

PURE PQC CERTIFICATES
HYBRID CERTIFICATES

TLS HYBRID KEY EXCHANGE SUPPORTED IN SOME STACKS

HYBRID IS REQUIRED FOR INTEROPERABILITY TODAY

LIMITATIONS

BROWSERS: NO WIDESPREAD NATIVE PQC TRUST

MANY LIBRARIES LACK SUPPORT

DEVICE/FIRMWARE CONSTRAINTS

STANDARDS AND IMPLEMENTATIONS STILL EVOLVING

Today = testing, preparation, and early phased adoption, not full production replacement

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The Operational Impact

PQC is an Operational Challenge

1. **Inventory cryptographic usage (certificates, keys, protocols)**
2. **Identify vulnerable algorithms (RSA/ECC dependencies)**
3. **Test PQC compatibility (applications, devices, libraries)**
4. **Orchestrate large-scale certificate replacement**
5. **Plan hybrid and phased migration strategies**

Identification of quantum-vulnerable algorithms is required for OMB M-23-02 reporting

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How Keyfactor Helps

Enabling PQC Readiness and Transition

DISCOVERY

OMB-aligned inventory and risk identification

Keyfactor AgileSec

Quantum readiness Assessments

Discover crypto assets and algorithm usage

Identify PQC exposure and prioritization

PKI

Hybrid and PQC issuance capability

Keyfactor EJBCA Enterprise

Supports PQC (pure, hybrid, composite)

Issue and test quantum safe certificates in controlled environments

Enables phased migration strategies

AUTOMATION

Large-scale certificate replacement

Keyfactor Command

Lifecycle automation at scale

Replace certificates across environments

Enforce policy during transition

EJBCA already supports PQC issuance today. The constraint is not capability, it's ecosystem readiness, as broader support across browsers, libraries, and platforms is still evolving.

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You Can't Manage What You Can't See

Federal Cryptographic Inventory & Discovery

Current Cryptographic Posture

- Scans your environment to identify certificates across public and private infrastructure
  - Internet-facing endpoints (public TLS)
  - Internal servers and applications (private PKI)
- Performs network-based discovery
  - Detects certificates in use on servers, load balancers, and endpoints
  - No prior knowledge or inventory required
- Uses domain and IP range scanning
  - Finds unknown or unmanaged certificates
  - Identifies shadow IT and forgotten assets

Supports OMB M-23-02 reporting requirements

Keyfactor AgileSec

- Integrates with cloud and platform APIs
  - Discovers certificates in cloud services (AWS, Azure, etc.)
  - Identifies certificates in modern, dynamic environments
- Collects detailed metadata to build context
  - Expiration dates
  - Issuing Certificate Authority
  - Key strength and configuration
  - Quantum readiness

Remediation Tracking

https://www.keyfactor.com/products/cryptographic-discovery-inventory/

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Enterprise PKI for Federal-Scale and Quantum-Ready Environments

Proven, Production-Scale PKI

- Trusted by global banks, telecom providers, and national governments
- Used in eID, ePassport, and national identity infrastructure
- Supports IoT deployments with tens of millions of devices
- Deployed in regulated environments (finance, healthcare, government)

Keyfactor
EJBCA
Enterprise

Flexible Deployment Models

- On-premises (data center, air-gapped environments)
- Cloud (AWS, Azure, federal cloud environments)
- Containerized (Docker images, Kubernetes via Helm)
- HSM-integrated for secure key storage (FIPS 140-2/3)

Quantum-Ready Today

- Issues pure and hybrid PQC certificates
- Supports ML-KEM, ML-DSA, and other emerging standards
- Enables controlled testing and phased migration

Standards-Based and Interoperable

- X.509, CMP, EST, ACME, SCEP
- Integrates with Microsoft ADCS and existing PKI environments
- Supports public and private trust models

Built for Cryptographic Agility

- Rapid algorithm changes without infrastructure redesign
- Supports hybrid transition strategies
- Future-proofed for evolving standards and mandates

EJBCA does not require immediate replacement of ADCS, it enables a phased transition from Windows-bound PKI to enterprise-scale, quantum-ready infrastructure.

https://www.keyfactor.com/products/ejbca-enterprise/

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Manage the Certificate Lifecycle

Monitor, Alert, and Automate

Keyfactor Command

- **Centralized certificate inventory**
  - Aggregates certificates across public and private environments
  - Provides a single system of record
- **Continuous monitoring**
  - Tracks certificate health and expiration in real time
  - Identifies risk before it becomes an outage
- **Proactive alerting**
  - Configurable notifications for upcoming expirations
  - Alerts based on policy, not just dates
- **One-click renewals**
  - Quickly replace certificates when needed
  - Reduces time to remediation during incidents
- **Automated certificate renewal**
  - Eliminates manual processes
  - Ensures certificates are renewed and deployed on time
- **Integration with public and private CAs**
  - Works with major public CAs (DigiCert, Let's Encrypt, etc.)
  - Supports internal/private PKI environments
- **Broad ecosystem integration**
  - Connects with load balancers, servers, cloud platforms, and DevOps tools
  - Enables automated deployment across infrastructure
- **Policy-driven management**
  - Enforces standards for certificate issuance and renewal
  - Enforces cryptographic policy aligned to NIST guidance (key size, algorithms, validity)

Keyfactor Command SaaS is FedRAMP 'In Process' and listed on the FedRAMP Marketplace

Command transforms certificate management from reactive firefighting into a controlled, automated process.

https://www.keyfactor.com/products/command/

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Bringing It All Together

From Certificate Agility to Crypto Agility

From certificate lifecycle automation to full cryptographic lifecycle management

Keyfactor AgileSec

Keyfactor Command

Keyfactor EJBCA Enterprise

Higher Frequency of Change

Greater Complexity of Change

Reduced Certificate Validity Periods

Expected Transition to Quantum Safe

Visibility
What do I have?

Automation
Can I scale?

Crypto Agility
Can I adapt?

The right cryptographic management platform moves you beyond solving challenges to managing cryptography as a strategic capability.

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Turning cryptographic complexity into recurring revenue

Partner Opportunity

Building a Cryptographic Services Practice

From point solutions to recurring cryptographic lifecycle services

Assess & Discover

Cryptographic Posture Assessments

Quantum Readiness Assessments

Cryptographic Inventory and Risk Analysis

Design & Implement

Cryptographic Strategy Development

Certificate Lifecycle Management Implementation

CA/PKI Modernization
(ADCS → EJBCA, hybrid models)

Operate & Optimize

Priority Certificate Remediation Services

Cryptographic Managed Services

Continuous Compliance and Policy Enforcement

AgileSec + Command + EJBCA enable a full cryptographic lifecycle services model

Keyfactor provides enablement across presales, implementation, and consulting delivery, helping partners build and scale a successful cryptographic services practice.

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Keyfactor University

Get certified today!

Delivering Value with Keyfactor Certification (v12)

Keyfactor Sales Training

Keyfactor Command 25.1 Overview

Keyfactor AgileSec Foundations

EJBCA Self-Service Training

Keyfactor certification equips partners with the technical credibility and practical expertise needed to confidently guide customers through PKI modernization, automation, and post-quantum readiness.

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Keyfactor Partner Center

Accelerate your business!

Register your deal!

Sales Enablement

Technical Resources

Marketing Assets

Product Information

The Keyfactor Partner Center provides everything partners need to sell, implement, and support Keyfactor solutions, including training, technical resources, sales tools, and deal support.

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Thank you

Mike Bodette
Senior Channel Solutions Engineer

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Common Questions cont.

1. “How disruptive is PQC really going to be?”

It’s significant, but it’s gradual. The disruption isn’t just new algorithms. It’s compatibility, performance, and large-scale replacement of cryptographic assets. That’s why it’s more of an operational challenge than just a technical one.

2. “When do we actually need to start worrying about PQC?”

Now. Not because quantum computers are breaking encryption today, but because of the “harvest now, decrypt later” risk. And also because standards are finalized and regulatory expectations are forming. The transition will take years, so waiting is the bigger risk.

3. “What’s the first step we should take?”

Start with visibility. You need to know where certificates and cryptography exist in your environment. From there, you can assess risk, prioritize, and begin automating lifecycle management. Without visibility, everything else is guesswork.

4. “Is this just a way for vendors to sell more tools?”

The drivers for this change are coming from the CA/Browser Forum and browser vendors, not from tooling vendors. The reality is that as lifetimes shrink and cryptography evolves, manual processes simply don’t scale. Tools aren’t the goal, they’re the enabler for managing what’s becoming a continuous process.

5. “Do we need to replace ADCS to adopt PQC?”

No. Most organizations start with a coexistence model. ADCS continues to support existing Windows-based workloads, while EJBCA is introduced for new use cases, higher scale environments, and PQC initiatives. Over time, organizations decide whether to reduce or fully replace ADCS based on their environment. This is a phased transition, not a rip-and-replace.

Common Questions cont.

6. “Can we use PQC in production today?”

Not broadly in production yet. Today, PQC is primarily used for testing, validation, and early-stage deployments. Hybrid certificates are the current approach, because the ecosystem, especially browsers and libraries, does not yet fully support pure PQC. Production adoption will increase as standards and support mature.

7. “What is the biggest blocker to PQC adoption today?”

The biggest blocker is ecosystem support. Browsers, operating systems, libraries, and devices are still catching up. That is why hybrid approaches are required today. The challenge is less about the algorithms themselves and more about compatibility across systems.

8. “How do we start preparing for PQC right now?”

Start with discovery. You need to identify where cryptography exists and where vulnerable algorithms like RSA and ECC are in use. From there, you can assess risk, prioritize systems, and begin testing PQC in controlled environments. Without visibility, you cannot plan a migration.

9. “What types of systems will be hardest to migrate to PQC?”

Systems with constrained environments or long lifecycles. That includes embedded devices, IoT systems, legacy applications, and anything with limited compute or firmware constraints. These systems are harder to update and often require more planning and testing.

10. “Will PQC replace TLS or just change what’s inside it?”

TLS itself does not go away. What changes are the cryptographic primitives inside it. Key exchange moves to KEM-based approaches like ML-KEM, and signatures move to PQC algorithms like ML-DSA. So, the protocol stays, but the underlying cryptography changes.

11. “How does PQC impact performance and infrastructure?”

PQC introduces larger keys and larger signatures, which increases handshake size and resource usage. That can impact CPU, memory, and network performance, especially in high-throughput environments. This is why testing and phased adoption are important.

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Common Questions cont.

12. “Why not wait until standards and tooling are fully mature?”

Because of the “harvest now, decrypt later” risk. Sensitive data being encrypted today could be stored and decrypted in the future once quantum capabilities mature. Also, migration takes years, not months. Waiting delays readiness and increases long-term risk.

13. “Where does Keyfactor fit compared to existing PKI tools?”

Keyfactor complements and extends existing PKI. It provides visibility, automation, and modern PKI capabilities that are required to manage scale, enforce policy, and support transitions like PQC. It is not just replacing tools, it is enabling capabilities that legacy systems were not designed for.

14. “Is PQC only relevant for national security systems?”

No. While CNSA 2.0 applies specifically to national security systems, the broader guidance from NIST, CISA, and OMB applies across federal environments. Any system handling sensitive or long-lived data should be planning for PQC.

Common Terms with Definitions cont.

1. “What is Post-Quantum Cryptography (PQC)?”

Post-quantum cryptography refers to cryptographic algorithms designed to remain secure against both classical and quantum computing attacks.

2. “What is RSA?”

RSA is a widely used public-key algorithm based on the difficulty of factoring large integers.

3. “What is ECC (Elliptic Curve Cryptography)?”

ECC is a public-key cryptography method based on solving discrete logarithm problems on elliptic curves.

4. “What is Shor’s Algorithm?”

Shor’s Algorithm is a quantum algorithm capable of efficiently solving problems like integer factorization and discrete logarithms, which would break RSA and ECC.

5. “What is ML-KEM (FIPS 203)?”

ML-KEM is a NIST-standardized post-quantum algorithm used for key establishment and encryption, based on lattice cryptography.

6. “What is ML-DSA (FIPS 204)?”

ML-DSA is a NIST-standardized post-quantum digital signature algorithm designed to replace RSA and ECDSA.

7. “What is SLH-DSA (FIPS 205)?”

SLH-DSA is a hash-based signature algorithm, meaning its security relies on well-understood hash functions rather than lattice problems. This makes it a more conservative option, but it comes with larger signatures and lower efficiency than alternatives like ML-DSA. As a result, it is typically used in high-assurance scenarios where long-term security is more important than performance, or where an alternative to lattice-based algorithms is preferred.

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Common Terms with Definitions cont.

8. “What is a hybrid certificate?”

A hybrid certificate includes both classical and post-quantum algorithms to maintain compatibility while transitioning to PQC.

9. “What is a KEM (Key Encapsulation Mechanism)?”

A KEM is a cryptographic method used to securely establish shared keys, replacing traditional key exchange methods like Diffie-Hellman.

10. “What is OMB M-23-02?”

OMB M-23-02 is a federal directive requiring agencies to inventory cryptographic systems, identify vulnerabilities, and plan for post-quantum migration.

11. “What is CNSA 2.0?”

CNSA 2.0 is an NSA framework that mandates the transition to quantum-resistant cryptographic algorithms for national security systems.

12. “What is NIST PQC standardization?”

NIST PQC standardization refers to the process of selecting and publishing quantum-resistant cryptographic algorithms as federal standards.

13. “What is FedRAMP?”

FedRAMP is a U.S. government program that standardizes security assessment and authorization for cloud services.

14. “What is ADCS?”

Active Directory Certificate Services (ADCS) is Microsoft’s PKI platform used to issue and manage certificates in Windows-based environments.

Common Terms with Definitions cont.

15. “What is Keyfactor EJBCA?”

EJBCA is a scalable, standards-based enterprise PKI platform used to issue and manage certificates across hybrid, cloud, and regulated environments.

16. “What is Keyfactor Command?”

Command is a certificate lifecycle automation platform that manages issuance, renewal, and deployment of certificates across environments.

17. “What is Keyfactor AgileSec?”

AgileSec is a cryptographic discovery and risk management platform that identifies certificates, keys, and algorithm usage across an enterprise.

18. “What is lattice cryptography?”

Lattice cryptography is based on working with points in a high-dimensional grid. The core problem is figuring out relationships between points in that space, even when there is noise or randomness added. Those problems are extremely difficult for both classical and quantum computers, which is why lattice-based algorithms are used in post-quantum cryptography.